Due Date

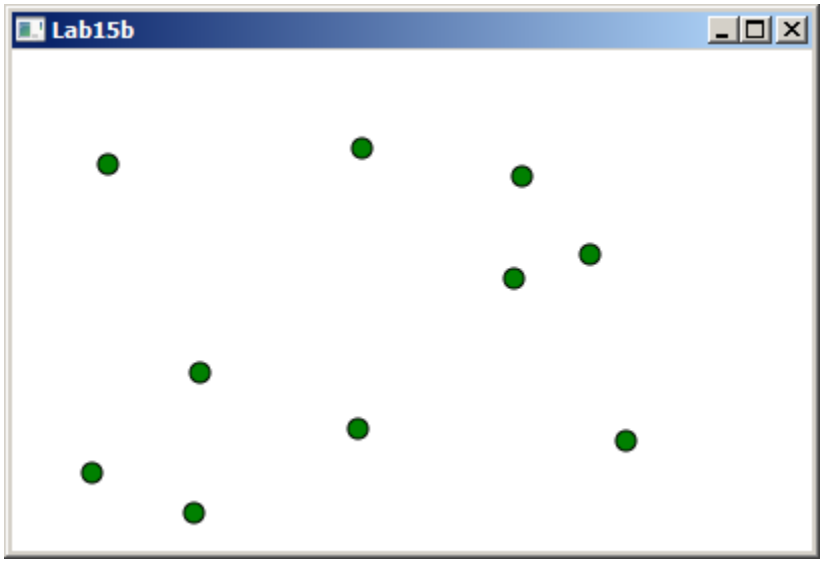
You must *submit* the source code for the solution to this lab exercise to *Moodle* by

Monday, November 24, 2025

in order to receive full credit for this work. You must also *demonstrate* the solution to the instructor during class, at the earliest opportunity.

Add / Remove Points on a JavaFX Display

Write a JavaFX application that lets the user click (the mouse) on a pane to create and remove "points". (Instead of actual points, use small circles.) When the user left-clicks the mouse (PRIMARY mouse button), a point is created and displayed at the current mouse coordinates. The user can then move the mouse to another position and left-click again to add another point at the new mouse position. In this way, the user can add as many points as they wish. If the user wishes to remove a point, they point to it with the mouse and right-click (SECONDARY mouse button).



A suitable way to draw the "points" is to represent each point with a **Circle** object having radius = 5 pixels. (If necessary, refer to the JavaFX API web site: https://openjfx.io/javadoc/21/

Remember that the list of nodes displayed in a JavaFX application can be accessed with code similar to the following:

```
ObservableList<Node> list = pane.getChildren();
for (int i = 0; i < list.size(); i++) {
    System.out.printf("i=%d, node=%s\n", i, list.get(i).toString());
}
```

Also remember that the methods to access the nodes in an **ObservableList** are similar to the methods you would use to access an **ArrayList**.

The online documentation for **ObservableList** is at:

https://openjfx.io/javadoc/21/javafx.base/javafx/collections/ObservableList.html

Optional Enhancement

If you wish, output a message to the console whenever the program responds to a left-click or right-click of the mouse, as shown in the example on the next page:

Optional Enhancement: console output

```
Lab15b EXECUTION STARTED.
LEFT-CLICK: adding point at x=162.0, y= 85.0
LEFT-CLICK: adding point at x=124.0, y=128.0
RIGHT-CLICK at mouse postion x=234.0, y=139.0:
RIGHT-CLICK at mouse postion x=238.0, y= 82.0:
RIGHT-CLICK at mouse postion x= 45.0, y=188.0:
RIGHT-CLICK at mouse postion x=120.0, y=125.0:
   Removing node 1 at x=124.0, y=128.0
RIGHT-CLICK at mouse postion x= 92.0, y=132.0:
RIGHT-CLICK at mouse postion x=111.0, y= 63.0:
RIGHT-CLICK at mouse postion x=159.0, y= 83.0:
   Removing node 0 at x=162.0, y= 85.0
```